

■ Study Session 4 — Multiple Regression & Model Evaluation

■ Goals

- Build and interpret a multiple linear regression model.
- Encode categorical predictors (e.g., gender, is_smoker) correctly.
- Evaluate fit with R^2 and Adjusted R^2 ; read residual plots.
- Detect issues (non-linearity, heteroscedasticity, multicollinearity).
- Save charts with `save_and_log_plot()` for reproducibility and indexing.

■ Steps

- 1 Load cleaned dataset (`data/processed/tips_cleaned.csv`) and inspect columns.
- 2 One-hot encode categoricals: `pd.get_dummies(df, columns=['gender','is_smoker'], drop_first=True)`.
- 3 Define `X = ['bill_total_usd','party_size','gender_Male','is_smoker_Yes']`, `y = tip_usd`.
- 4 Fit `LinearRegression()`; print coefficients, intercept, and R^2 .
- 5 Compute Adjusted R^2 : $1 - (1 - R^2) * (n - 1) / (n - p - 1)$ (n =rows, p =#features).
- 6 Plot Actual vs Predicted with red dashed $y=x$ line; save via `save_and_log_plot()`.
- 7 Plot residuals ($y - y_{\text{pred}}$) vs predicted; add horizontal line at 0.
- 8 Optional: check multicollinearity (VIF) and drop/transform if needed.
- 9 Write a short Markdown interpretation: which variables matter and why?

■ Multicollinearity (VIF) — Optional

- VIF quick check (optional): for each numeric column in `X`:
- $VIF \approx 1 / (1 - R^2_{\text{from regressing that feature on all others}})$.
- High VIF (> 5 – 10) suggests multicollinearity; consider dropping or combining.

■ Reflection Questions

- Did adding predictors improve Adjusted R^2 meaningfully?
- Which coefficients were strongest, and do they match intuition?
- Do residuals look randomly scattered (good) or patterned (bad)?
- Any signs of heteroscedasticity (funnel shape) or outliers?

■ ELI5 Summary

Multiple regression is like guessing tips using several clues at once: bill size, group size, and whether someone smokes. Each coefficient says how much tips change when that one clue changes, while the others stay the same. The red dashed line ($y = x$) shows perfect guesses—your points getting close to it means your model is doing well.